

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name: Ashwin Varshney

Enrollment No. NJG241435

Jaypee Institute of Information Technology, Noida

TI Examination, ODD Semester 2024

B. Tech, 1st Semester

Course Title: Basic Electronics

Max Time: 1 Hour

Course Code: 24B11EC111

Max Marks: 20

CO1	Recall the concepts of various circuit elements and Kirchhoff's laws.
CO2	Understand the basics of semiconductor p-n junction diodes, op-amps and their applications.
CO3	Apply network theorems to solve complex DC networks.
CO4	Explain the operations of transistors (BJT and MOSFET) and analyze their biasing techniques.

Note: Attempt all the questions.

Q.1. Using Kirchhoff's laws, determine the values of i_4 , i_2 , v_3 and v_6 in figure 1.

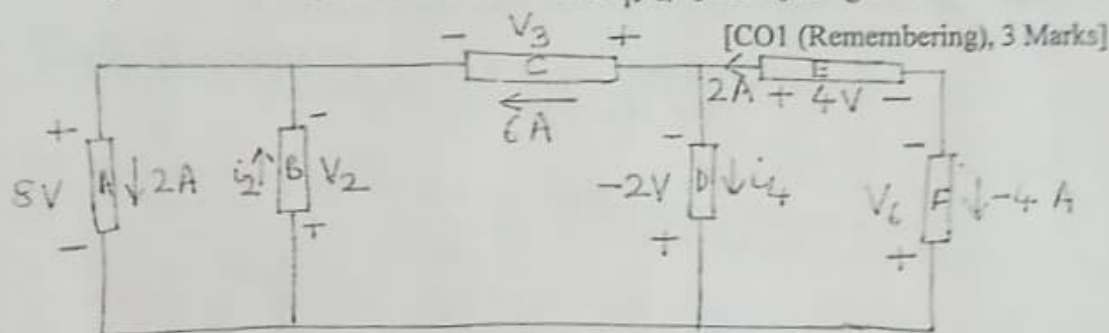


Figure 1.

Q.2. Use current division to determine the currents i_1 and i_4 in the circuit shown in figure 2.

[CO1 (Remembering), 2 Marks]

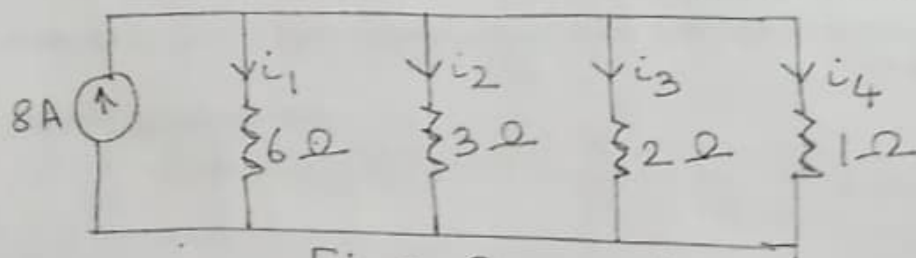


Figure 2.

Q.3. Find the branch currents i_1 , i_2 and i_3 using mesh analysis in figure 3.

[CO1 (Remembering), 3 Marks]

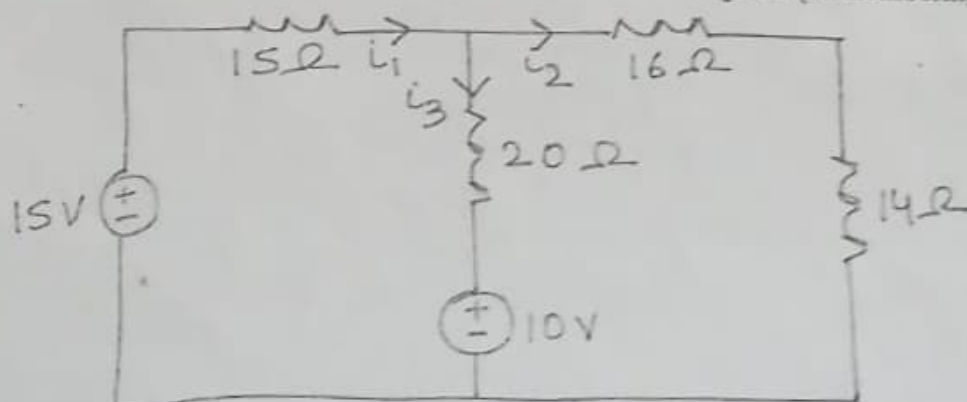


Figure 3.

Q.4. Determine node voltages (v_1 , v_2) for the circuit shown in figure 4.

[CO1 (Remembering), 3 Marks]

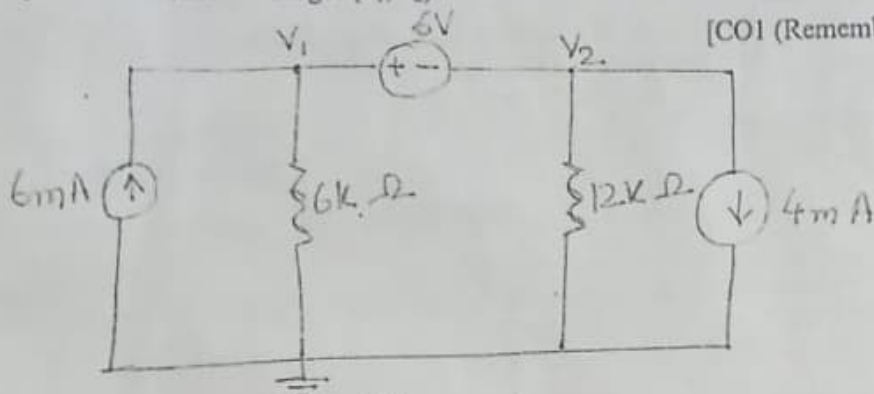


Figure 4.

Q.5. Use superposition theorem to find 'i' for the circuit shown in figure 5.

[CO3 (Applying), 4 Marks]

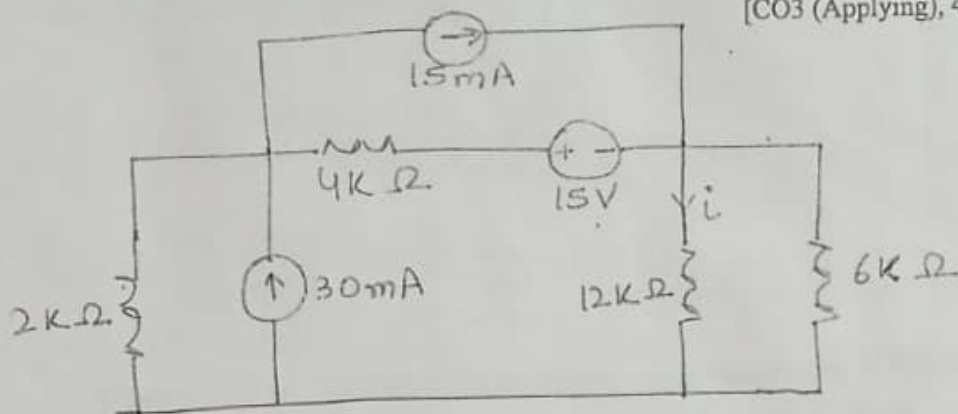


Figure 5

Q.6. Determine thevenin's equivalent circuit between terminal a and b of the circuit shown in figure 6 and find v_0 .

[CO3 (Applying), 5 Marks]

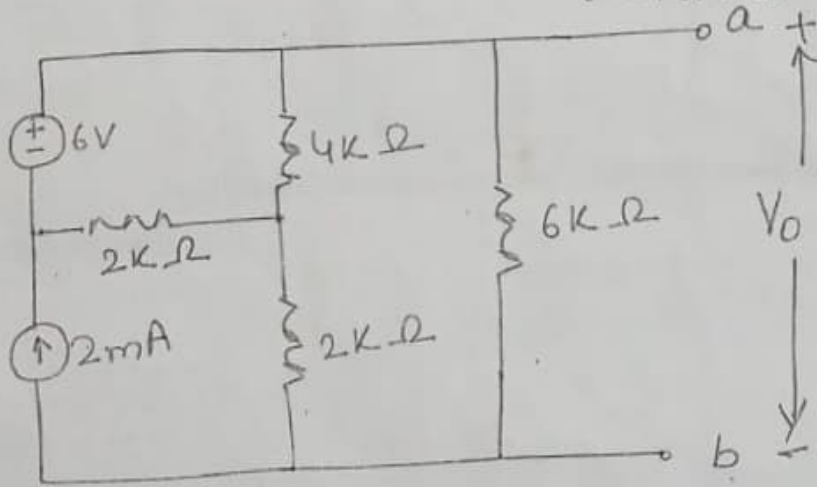


Figure 6.

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name Aayon Vansh

Enrollment No. 992401030154

Jaypee Institute of Information Technology, Noida
Test-2 Examination, ODD Semester 2024
B.Tech, I Semester

Course Title: Basic Electronics
Course Code: 24B11EC111

Max Time: 1 Hour
Max Marks: 20

Course Outcomes: At the end of the course, students will be able to:

CO1	Recall the concepts of various circuit elements and Kirchhoff's laws.
CO2	Understand the basics of semiconductor p-n junction diodes, op-amps and their applications.
CO3	Apply networks theorems to solve the complex DC networks.
CO4	Explain the operations of transistors (BJT and MOSFET) and analyze their biasing techniques.

Note: Attempt all the questions.

Q.1. (i). The current flowing through a p-n junction diode at room temperature is 100 nA when a large reverse bias voltage is applied. Calculate the current through the diode when 0.5 V is applied in the forward bias mode. Given $k = 1.38 \times 10^{-23} \text{ J/K}$, $T = 300\text{K}$.

[CO2 (Understanding), 2 Marks]

(ii). Derive an expression for efficiency of half wave rectifier.

[CO2 (Understanding), 2 Marks]

Q.2. A full wave rectifier is supplied with a sinusoidal voltage of 220 V at 50 Hz. The transformer connected to the rectifier has a turn ratio of $N_1 : N_2 = 10:1$. Assuming all the diodes to be ideal, determine the following:

- DC output voltage.
- R.M.S output voltage.
- Form factor.
- Ripple factor.

[CO2 (Understanding), 1+1+1+1 Marks]

Q.3. Sketch the waveform for V_o for the equilibrium state of the circuit shown in figure 1. Consider the diode as Silicon diode ($V_T = 0.7 \text{ V}$). Solve it with the help of required steps.

[CO2 (Understanding), 4 Marks]

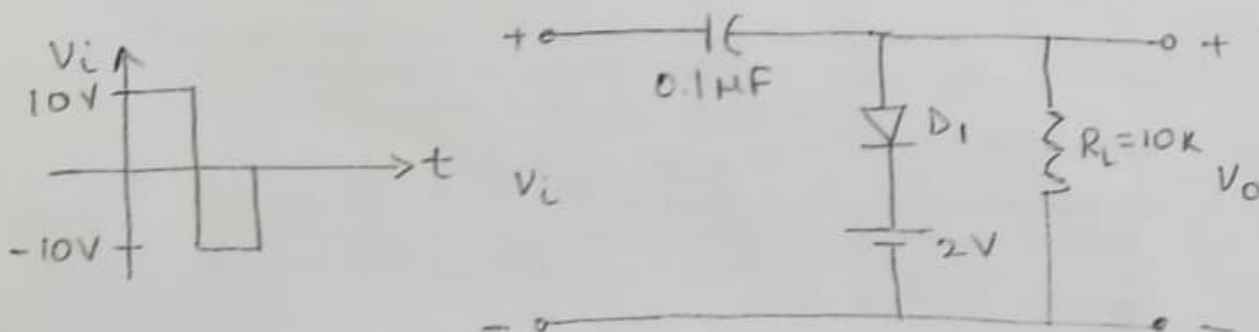


Figure 1

Q.4. Determine V_o for the circuit shown in figure 2. Consider the diode as Silicon diode ($V_T = 0.7$ V).

[CO2 (Understanding), 4 Marks]

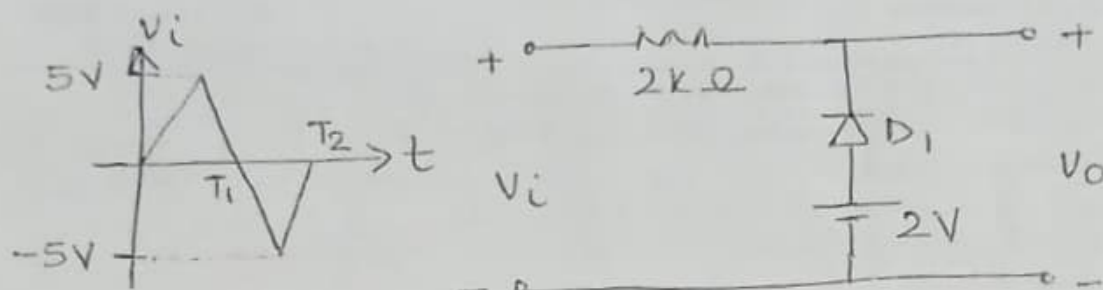


Figure 2.

Q.5. For the circuit shown in figure 3, determine the range of R_L and I_L that will maintain V_{RL} at 20 V. Also, calculate the maximum wattage rating required for the zener diode of the circuit.

[CO3 (Applying), 4 Marks]

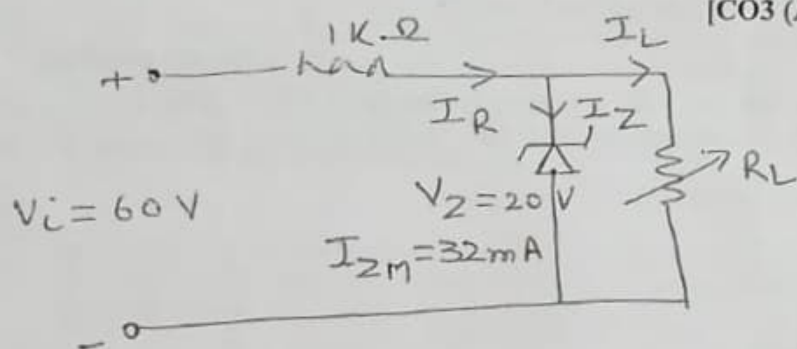


Figure 3

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POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Aayon Varshney

Enrollment No. 992401030154

Jaypee Institute of Information Technology, Noida
End Term Examination, ODD Semester 2024
B.Tech./IMT 1st Semester

Course Title: Basic Electronics
Course Code: 24B11EC111

Maximum Time: 2 Hrs
Maximum Marks: 35

CO1	Recall the concepts of various circuit elements and Kirchhoff's laws.
CO2	Understand the basics of semiconductor PN junction diodes and Op-Amp, and their applications.
CO3	Apply network theorems to effectively solve complex DC circuits.
CO4	Explain the operation of transistors (BJT and MOSFET) and analyze their biasing techniques.

Note: Attempt all the questions

- Q1. The voltages V_a , V_b and V_c in Fig. 1, are the node voltages corresponding to nodes a, b and c respectively. Find the value of 'A', the gain of the dependent source.

[2 Marks, CO1 (Remembering)]

- Q2. (a) Apply the Mesh analysis to find all the mesh currents in Fig. 2. Given $i_3 = 3.93$ A.
(b) Draw the Thevenin's equivalent circuit for the network shown in Fig. 3.

[4+3 Marks, CO3 (Applying)]

- Q3. (a) Differentiate between junction capacitance under forward bias and reverse bias.
(b) Compute the current I_1 , I_2 and I_3 for the network shown in Fig. 4. Use simplified model for the diodes. Assume diode is silicon diode.
(c) For the network of Fig. 5 (b), (i) Calculate 5 time constant (5τ) (ii) Illustrate the output voltage V_o for the waveform shown in Fig. 5(a).

[2+2+3 Marks, CO2 (Understanding)]

- Q4. The Zener diode in the regulator circuit shown in Fig. 6 has a Zener voltage of 5.8 volts and a Zener knee current (I_{zmin}) of 0.5 mA. Calculate the maximum load current ensuring proper functioning over the input voltage range between 20 and 30 volts.

[3 Marks, CO3 (Applying)]

- Q5. (a) Explain the base width modulation in common emitter configuration of BJT.
(b) For a Common Emitter circuit shown in Fig. 7, $V_{CC} = 10$ V, $\beta = 100$, $V_{BE} = 0.7$ V, $I_{CQ} = 10$ mA, $V_{CE} = 5$ V Find the resistance values (R_B , R_C).
(c) For the circuit shown in Fig. 8, find the stability factor when $R_E = 1.2$ k Ω and then changed to $R_E = 2.2$ k Ω . Compare the two values of emitter bias resistance. Which is better for better stability. Assume $\beta = 50$.

[2+2+3 Marks, CO4 (Analysing)]

- Q6. (a) What is pinch off condition in MOSFET. Explain with the help of a neat diagram.
(b) Find $V_{DS(min)}$ needed to operate MOSFET in saturation region. $I_D = 200$ μ A, $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$, $t_{ox} = 8$ nm, $\mu_n = 450$ cm²/V-s, $\epsilon_{ox} = 3.9\epsilon_0$, $V_{th} = 0.7$ V.

[2+2 Marks, CO4 (Analysing)]

- Q7. (a) Discuss the characteristics of an ideal Op-Amp.
(b) If sinewave of peak amplitude 2V and frequency 1 kHz is applied at the input of an ideal differentiator with $C_1 = 0.1$ μ F and $R_1 = 1$ K Ω , then write the expression of the output voltage. Also illustrate output and input waveforms marking voltage levels.

[2+3 Marks, CO2 (Understanding)]

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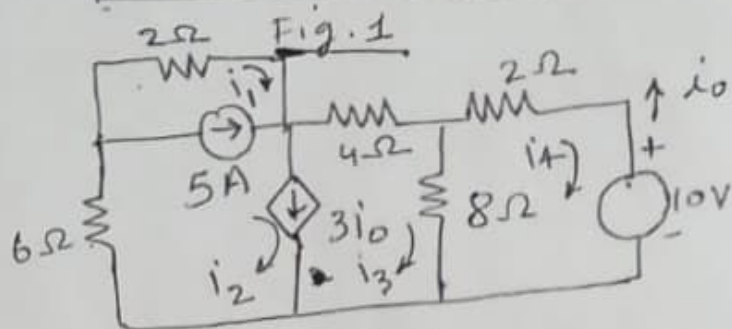
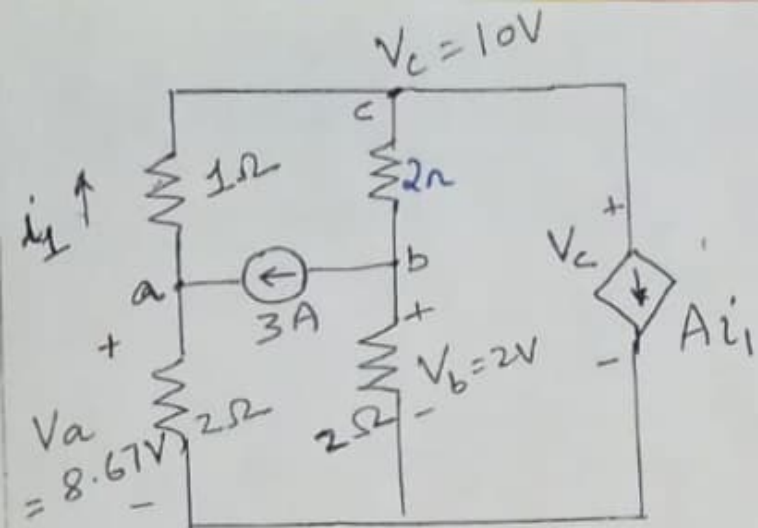


Fig. 2

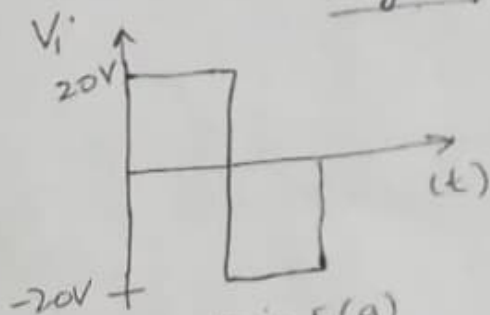


Fig 5(a)

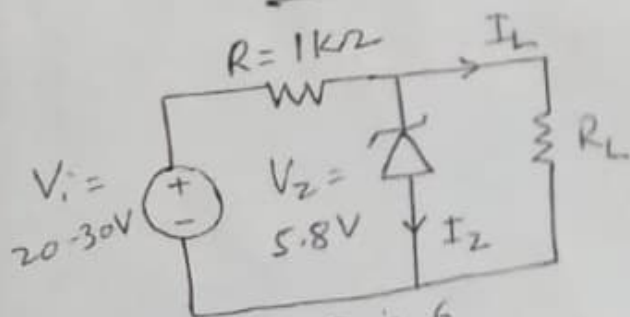


Fig. 6

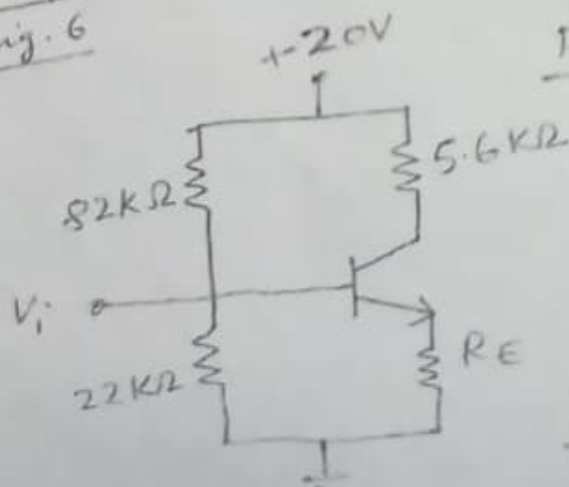


Fig 8

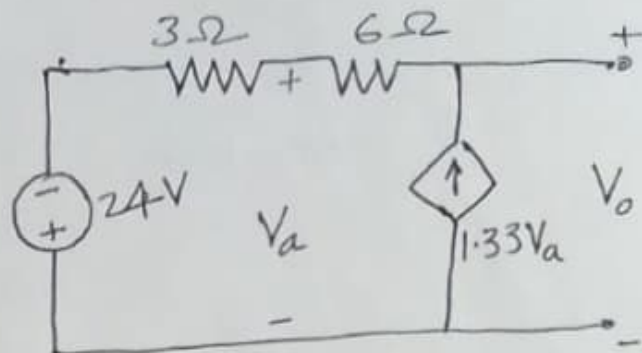


Fig 3:

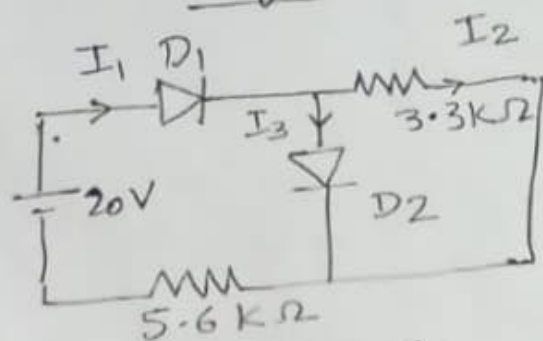


Fig. 4

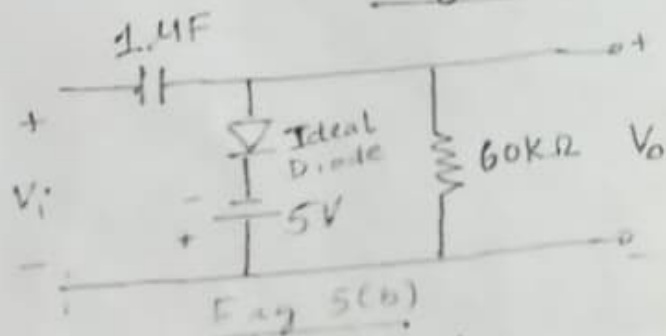


Fig 5(b)

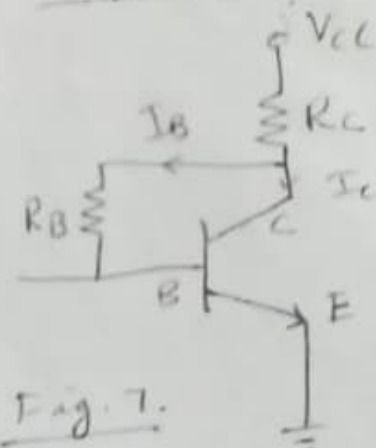


Fig. 7.